

Uniform Diophantine approximation and run-length function in continued fractions

Zhou Qing-Long

Wuhan University of Technology

This is a joint work with Bo Tan

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- 1 Asymptotic/Uniform Diophantine Approximation
- 2 Continued Fraction and Pre-dimensional number
- 3 Applying: Run-length function
- 4 Ideas and proofs

Abstract

We study the multifractal properties of the asymptotic/uniform approximation exponent

$$\nu(x) = \sup\{\nu \geq 0: |T^n(x) - y| < |I_n(y)|^\nu \text{ for i.m. } n\},$$

$$\hat{\nu}(x) = \sup\{\hat{\nu} \geq 0: \forall N \gg 1, \exists n \in [1, N], \text{ s.t. } |T^n(x) - y| < |I_N(y)|^{\hat{\nu}}\}.$$

As a corollary, we determine the Hausdorff dimension of the uniform Diophantine set

$$\{x \in [0, 1): \forall N \gg 1, \exists n \in [1, N], \text{ s.t. } |T^n(x) - y| < |I_N(y)|^{\hat{\nu}}\}$$

for algebraic irrational points $y \in [0, 1)$.

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Metric Diophantine approximation

Theorem 1 (Dirichlet's Thm, 1842)

For any $x \in \mathbb{R}$ and $N > 1$, there exists integers n such that

$$||nx|| \leq \frac{1}{N} \text{ and } 1 \leq n < N,$$

where $|| \cdot ||$ denotes the distance to the nearest integer.

Dirichlet Thm provides a **uniform** estimate of $||nx||$ in terms of N .
This kind of approximation is referred to uniform approximation.

Corollary 2

$$\{x \in \mathbb{R} : ||nx|| < n^{-1} \text{ for i.m. } n\} = \mathbb{R}$$

- This kind of approximation is referred to **asymptotic** approximation.
- In Dirichlet's corollary, the approximation function $1/n$ can only be improved to $1/\sqrt{5}n$ (Hurwitz's Theorem).
- Main reason: bad points (eg: $x = \frac{\sqrt{5}-1}{2} = [1, 1, \dots,]$).

Metric Diophantine approximation.

Instead of considering **all points**,
consider the properties of **generic points** (in a sense of measure).

Asymptotic Diophantine approximation

Let $\psi : \mathbb{N} \rightarrow \mathbb{R}^+$. The set of ψ -well approximable points:

$$\mathcal{W}(\psi) := \left\{ x \in [0, 1) : \|nx\| < \psi(n), \text{ i.m. } n \right\}$$

How about the size of $\mathcal{W}(\psi)$ in the sense of
→ Lebesgue measure and fractal dimensions?

Size in Lebesgue measure

Theorem 3 (Khintchine's theorem, 1924)

Let $\psi : \mathbb{N} \rightarrow \mathbb{R}^+$ be *decreasing*. Then

$$\mathcal{L}(\mathcal{W}(\psi)) = 0 \text{ or } 1 \iff \sum_{q=1}^{\infty} \psi(q) < +\infty \text{ or } = +\infty.$$

- The convergence part of Khintchine is a direct result of Borel-Cantelli Lemma; it remains true even without the monotonicity of ψ .
- Duffin & Schaeffer (1941) constructed a counterexample to show that monotonicity assumption is essential in the divergence part.

Monotonicity in Khintchine

For $\psi : \mathbb{N} \rightarrow [0, \frac{1}{2})$,

$$\mathcal{W}^*(\psi) := \left\{ x \in [0, 1) : |nx - m| < \psi(n), \gcd(n, m) = 1 \text{ i.o.} \right\}$$

Remark: $\mathcal{W}^*(\psi) = \mathcal{W}(\psi)$ if ψ is decreasing.

Duffine-Schaeffer Conjecture (1941):

$$\mathcal{L}(\mathcal{W}^*(\psi)) = 0 \text{ or } 1 \iff \sum_{n=1}^{\infty} \frac{\phi(n)}{n} \psi(n) < +\infty \text{ or } = +\infty.$$

The proof of [Koukoulopoulos-Maynard 2020](#).

A quantitative version [Aistleitner-Borda-Hauke 2023](#).

Size in Hausdorff dimension

Theorem 4 (Jarník theorem, 1931)

Let $\psi : \mathbb{N} \rightarrow \mathbb{R}^+$ be *decreasing*. Then

$$\dim_H \mathcal{W}(\psi) = \frac{2}{\tau + 1}, \quad \text{where } \tau = \liminf_{q \rightarrow \infty} \frac{-\log \psi(q)}{\log q}.$$

■ Khintchine Theorem $\xRightarrow{\text{MTP with } f(x)=x^{2/\tau+1}}$ Jarník Theorem

Mass Transference Principle

■ Beresnevitch, Velani 2006

Let $\{B_i\}_{i \in \mathbb{N}}$ be a sequence of balls in $\mathbb{R}^k$ with $r(B_i) \rightarrow 0$ as $i \rightarrow \infty$. Let f be a dimension function such that $x^{-k}f(x)$ is monotonic and suppose that for any ball $B \subset \mathbb{R}^k$

$$\mathcal{H}^k(B \bigcap \limsup_{i \rightarrow \infty} B_i^f) = \mathcal{H}^k(B).$$

Then, for any ball B in $\mathbb{R}^k$

$$\mathcal{H}^f(B \bigcap \limsup_{i \rightarrow \infty} B_i^k) = \mathcal{H}^f(B).$$

Asymptotic approximation: Shrinking target problem

Shrinking target problem (Hill-Velani 1995): let (X, d) be a metric space and $T: X \rightarrow X$ be a transformation. Let $\psi: \mathbb{N} \rightarrow \mathbb{R}^+$ be a decreasing function.

How about the size of

$$\begin{aligned} W(y, \psi) &:= \{x \in X : d(T^n x, y) < \psi(n) \text{ i.m. } n\} \\ &= \{x \in X : T^n x \in B(y, \psi(n)) \text{ i.m. } n\} \\ &= \limsup_{n \rightarrow \infty} T^{-n} B(y, \psi(n)) \end{aligned}$$

→ Lebesgue measure and fractal dimensions?

Some measure results

Borel-Cantelli Lemma Let μ be an invariant measure. Then

$$\sum_{n \geq 1} \mu(B(y, \psi(n))) < \infty \Rightarrow \mu(W(y, \psi)) = 0.$$

Dynamical Borel-Cantelli Lemma

$$\sum_{n \geq 1} \mu(B(y, \psi(n))) = \infty + \text{some condition} \Rightarrow \mu(W(y, \psi)) = \text{full}.$$

- **Philipp 1967**: β -transformation, Gauss transformation.
- **Kleinbock-Margulis 1999**: dynamics on homogeneous spaces.
- **Chernov-Kleinbock 2001**: Anosov diffeomorphisms and topological Markov chains, Gibbs measure.

Some dimension results

- Hill-Velani 1995, 1997: expanding rational map on Julia set.
- Hill-Velani 1999: linear maps on n -dimensional tori.
- Li-Liao-Velani-Zorin 2022: real matrix Transformations of tori.
- Urbański 2002: conformal IFS.
- Allen-Bárány 2021: Hausdorff measure, conformal IFS.
- Shen-Wang 2013: β -transformation.
- Coons-Hussain-Wang 2016: Hausdorff measure, β -transfor.
- Li-Wang-Wu-Xu 2014: continued fractions.

Uniform Diophantine Approximation

Hitting problem (Kelmer 2017): Let (X, T) be a exponentially mixing system with respect to the probability measure μ . Let $\psi: \mathbb{N} \rightarrow \mathbb{R}^+$ be a decreasing function. Fix a point $y \in X$.

- $O_N(x) := \{Tx, T^2x \dots, T^Nx\};$
- Target $B_N := B(y, \psi(N));$
- $E_N := \{x \in X: O_N(x) \cap B_N = \emptyset\}.$

How about the size of

$$\begin{aligned} \mathcal{U}(y, \psi) &:= \{x \in X: O_N(x) \cap B_N \neq \emptyset \text{ for } N \text{ ultimately} \} \\ &= X \setminus \limsup_{N \rightarrow \infty} E_N = X \setminus \bigcap_{m \in \mathbb{N}} \bigcup_{n \geq m} E_m \end{aligned}$$

→ Lebesgue measure and fractal dimensions?

Size in Lebesgue measure

Theorem 5 (Kleinbock-Konstantoulas-Richter, 2019)

Let $(X, \mu, T, \mathcal{B})$ be a exponentially mixing system. Let $\psi : \mathbb{N} \rightarrow \mathbb{R}^+$ be *decreasing*.

$$\mu(\mathcal{U}(y, \psi)) = \begin{cases} \text{null,} & \text{if } \sum_{n=1}^{\infty} \frac{\mu(E_n)^{1-\varepsilon}}{n} < \infty \text{ for some } \varepsilon > 0; \\ \text{full,} & \text{if } \sum_{n=1}^{\infty} \frac{\mu(E_n)}{n} = \infty. \end{cases}$$

- T : β -transformation, Gauss transformation.
- **Further explorations**: Is it possible to remove ε ? It remains true even without the monotonicity of ψ .

Some known measure results

- [Kelmer 2017](#): hyperbolic manifold of finite volume.
- [Kelmer-Oh 2018](#): hyperbolic manifold of infinite volume.
- [Kelmer-Yu 2019](#): higher rank homogeneous spaces
- [Kirsebom-Kund-Persson 2020](#): some classes of interval maps (doubling map, Gauss map, Manneville-Pomeau map)
- [Ganotaki-Persson 2021](#): Impose a slightly stronger conditions.
Motivation: In order to conclude a zero-one law.

Size in Hausdorff dimension

Definition 6 (Amou-Bugeaud, 2010)

For $x \in [0, 1]$, we define the asymptotic approximation exponent of x by

$$\nu(x) = \sup \left\{ \nu \geq 0 : |T^n(x) - y| < |I_n(y)|^\nu \text{ i.m. } n \right\}$$

and the uniform approximation exponent by

$$\hat{\nu}(x) = \sup \left\{ \hat{\nu} \geq 0 : \forall N \gg 1, \exists n \in [1, N], \text{ s.t. } |T^n(x) - y| < |I_N(y)|^{\hat{\nu}} \right\}.$$

We define two exceptional sets by

- $E(\hat{\nu}) := \{x \in [0, 1) : \hat{\nu}(x) = \hat{\nu}\};$
- $E(\hat{\nu}, \nu) := \{x \in [0, 1) : \hat{\nu}(x) = \hat{\nu}, \nu(x) = \nu\}.$

Dimension results: β -dynamical system

Theorem 7 (Bugeaud-Liao, 2016)

For β -transformation $T_\beta : x \rightarrow \beta x \bmod 1$ on $[0,1)$. Given two nonnegative real numbers $\hat{\nu}$ and ν with $\hat{\nu} \leq \nu$, if $\hat{\nu} > \frac{\nu}{1+\nu}$, then $E(\hat{\nu}, \nu)$ is empty. Otherwise, we have

$$\dim_H E(\hat{\nu}, \nu) = \frac{\nu - 1 - \nu\hat{\nu}}{(1 + \nu)(\nu - \hat{\nu})}.$$

Theorem 8 (Bugeaud-Liao, 2016)

Let $\hat{\nu}$ be a real number in $[0,1]$. Then

$$\dim_H E(\hat{\nu}) = \dim_H \mathcal{U}(0, \beta^{-n}) = \left(\frac{1 - \hat{\nu}}{1 + \hat{\nu}} \right)^2.$$

- Wu 2020: extend the result of B-L to every point $y \in [0, 1]$.
- Wu-Zheng 2022: for a general approximation function ψ .
- Wang-Li-Li 2023: β is an integer. $T_\beta^n \xrightarrow{\text{is replaced by}} T_\beta^{a_n}$,
where $a_n \uparrow$ of positive integers with $\limsup_n \frac{a_{n+1}}{a_n} < +\infty$.

Question: for continued fraction system.

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Continued fraction expansion

$$\forall x \in (0, 1),$$

$$\begin{aligned}
 x &= \frac{1}{\frac{1}{x}} = \frac{1}{\lfloor \frac{1}{x} \rfloor + \{\frac{1}{x}\}} \\
 &= \frac{1}{\lfloor \frac{1}{x} \rfloor + \frac{1}{\frac{1}{\{\frac{1}{x}\}}}} = \frac{1}{\lfloor \frac{1}{x} \rfloor + \frac{1}{\lfloor \frac{1}{\{\frac{1}{x}\}} \rfloor + \{\frac{1}{\{\frac{1}{x}\}}\}}} \\
 &= \dots \\
 &= \frac{1}{a_1(x) + \frac{1}{a_2(x) + \frac{1}{\ddots}}}
 \end{aligned}$$

Partial quotient and convergent

Continued fraction expansion of x :

$$\begin{aligned}
 x &= \frac{1}{a_1(x) + \frac{1}{a_2(x) + \frac{1}{a_3(x) + \cdots}}} \\
 &= [a_1(x), a_2(x), a_3(x), \cdots].
 \end{aligned}$$

- Partial quotient: $a_n(x) = \left\lfloor \frac{1}{T^{n-1}(x)} \right\rfloor$.
- Convergent: $\frac{p_n(x)}{q_n(x)} = [a_1(x), \dots, a_n(x)]$.

Continued fractions

$$p_{-1}(x) = 1, \quad p_0(x) = 0, \quad p_n(x) = a_n(x)p_{n-1}(x) + p_{n-2}(x);$$

$$q_{-1}(x) = 0, \quad q_0(x) = 1, \quad q_n(x) = a_n(x)q_{n-1}(x) + q_{n-2}(x).$$

Definition 9 (Khintchine, 1963)

For $n \geq 1$ and $(a_1, \dots, a_n) \in \mathbb{N}^n$, we call

$$I_n(a_1, \dots, a_n) = \{x \in [0, 1) : a_k(x) = a_k, 1 \leq k \leq n\}$$

a cylinder of order n .

- $I_n(x) := I_n(a_1(x), \dots, a_n(x)).$

Pre-dimensional number

Let $\mathcal{A} \subseteq \mathbb{N}$ be a finite or infinite subset and $B > 1$ be a fixed number. For any $n \geq 1$ and $\rho \geq 0$, define

$$f_n(\rho) = \sum_{a_1, \dots, a_n \in \mathcal{A}} \frac{1}{(B^n q_n^2)^\rho}.$$

- Pre-dimensional number (Wang-Wu, 2008):

$$s_{n,B}(\mathcal{A}) = \inf\{\rho \geq 0: f_n(\rho) \leq 1\}.$$

Lemma 10 (Wang-Wu, 2008)

- (1) *The limit $\lim_n s_{n,B}(\mathcal{A})$ exists, denoted by $s_B(\mathcal{A})$.*
 (2) *For any $\alpha \in \mathbb{N}$, take $\mathcal{A}_\alpha = \{1, 2, \dots, \alpha\}$. Then*

$$\lim_{\alpha \rightarrow \infty} s_B(\mathcal{A}_\alpha) = s_B(\mathbb{N}) := s_B.$$

- (3) *s_B is continuous with respect to $B \in (1, \infty)$. Furthermore,*

$$\lim_{B \rightarrow 1} s_B = 1, \quad \lim_{B \rightarrow \infty} s_B = \frac{1}{2}.$$

Remark: s_B can be also given as the solution to the pressure function

$$P(-s(\log |T'| + \log B)) = 0,$$

where $P(\phi)$ is defined as

$$P(\phi) := \lim_{n \rightarrow \infty} \frac{\log \sum_{y: T^n y = x} \exp\{\phi(y) + \dots + \phi(T^{n-1}y)\}}{n}.$$

Main results

Consider a class of quadratic irrational numbers $y = [i, i, \dots]$ with $i \in \mathbb{N}$. Write $f(a, b) = a^{\frac{b}{1-b}}$.

Theorem 11 (Tan-Z, 2023)

Given two nonnegative real numbers $\hat{\nu}$ and ν with $\hat{\nu} \leq \nu$, we have

$$\dim_H E(\hat{\nu}, \nu) = \begin{cases} 1, & \text{if } \nu = 0; \\ s_{B_1}, & \text{if } 0 \leq \hat{\nu} \leq \frac{\nu}{1+\nu} < \nu \leq \infty; \\ 0, & \text{otherwise,} \end{cases}$$

where $B_1 = f\left(\frac{i+\sqrt{i^2+4}}{2}, \frac{\nu^2}{(1+\nu)(\nu-\hat{\nu})}\right)$.

Main results

Theorem 12 (Tan-Z, 2023)

Given a nonnegative real number $\hat{\nu}$, we have

$$\dim_H E(\hat{\nu}) = \dim_H \mathcal{U}(y, \hat{\nu}) = \begin{cases} s_{B_2}, & \text{if } 0 \leq \hat{\nu} \leq 1; \\ 0, & \text{otherwise,} \end{cases}$$

where $B_2 = f\left(\frac{i+\sqrt{i^2+4}}{2}, \frac{4\hat{\nu}}{(1+\hat{\nu})^2}\right)$.

Remark: These results remain valid for any quadratic irrational number

$$y = [a_1, a_2, \dots, a_{k_0}, \overline{a_{k_0+1}, \dots, a_{k_0+h}}],$$

where $k_0, h \in \mathbb{N}$.

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Run-length function: Dyadic system

For $x \in [0, 1]$ with dyadic expansion

$$x = \sum_{n=1}^{\infty} \frac{x_n}{2^n} \quad (x_n = 0 \text{ or } 1).$$

We define the run-length function $r_n(x)$ as

$$r_n(x) = \max \{l \geq 1 : x_{i+1} = \cdots = x_{i+l} = 0 \text{ for some } 0 \leq i \leq n-l\}.$$

Theorem 13 (Erdős-Rényi, 1970)

For Lebesgue almost all $x \in [0, 1]$,

$$\lim_{n \rightarrow \infty} \frac{r_n(x)}{\log_2 n} = 1.$$

Run-length function: Continued fraction system

For $n \geq 1$, we define the n -th maximal run-length function of x as

$$R_n(x) = \max \{l \geq 1 : a_{i+1}(x) = \cdots = a_{i+l}(x) \text{ for some } 0 \leq i \leq n-l\}.$$

Theorem 14 (Wang-Wu, 2011)

For Lebesgue almost all $x \in [0, 1]$,

$$\lim_{n \rightarrow \infty} \frac{R_n(x)}{\log_{\frac{\sqrt{5}+1}{2}} n} = \frac{1}{2}.$$

Run-length function: Continued fraction system

Size in Hausdorff dimension

Let $\varphi: \mathbb{N} \rightarrow \mathbb{R}^+$ be a non-decreasing function. Define

$$E(\{\varphi(n)\}_{n=1}^{\infty}) = \left\{ x \in [0, 1) : \lim_{n \rightarrow \infty} \frac{R_n(x)}{\varphi(n)} = 1 \right\},$$

$$F(\{\varphi(n)\}_{n=1}^{\infty}) = \left\{ x \in [0, 1) : \limsup_{n \rightarrow \infty} \frac{R_n(x)}{\varphi(n)} = 1 \right\},$$

- (Wang-Wu 2011) case 1: $\lim_n \frac{\varphi(n+\varphi(n))}{\varphi(n)} = 1$;
 case 2: $\liminf_n \frac{\varphi(n)}{n} = \beta \in [0, 1]$.

We complete the study by considering the sets

$$E_\alpha = \left\{ x \in [0, 1) : \liminf_{n \rightarrow \infty} \frac{R_n(x)}{n} = \alpha \right\},$$

$$E_{\alpha, \beta} = \left\{ x \in [0, 1) : \liminf_{n \rightarrow \infty} \frac{R_n(x)}{n} = \alpha, \limsup_{n \rightarrow \infty} \frac{R_n(x)}{n} = \beta \right\}.$$

Theorem 15 (Tan-Z, 2023)

For $\alpha, \beta \in [0, 1]$ with $\alpha \leq \beta$, we have

$$\dim_H E_{\alpha, \beta} = \begin{cases} 1, & \text{if } \beta = 0, \\ s_{B_3}, & \text{if } 0 \leq \alpha \leq \frac{\beta}{1+\beta} < \beta \leq 1, \\ 0, & \text{otherwise,} \end{cases}$$

where $B_3 = f\left(\frac{\sqrt{5}+1}{2}, \frac{\beta^2(1-\alpha)}{\beta-\alpha}\right)$.

Theorem 16 (Tan-Z, 2023)

For $\alpha \in [0, 1]$, we have

$$\dim_H E_\alpha = \begin{cases} s_{B_4}, & \text{if } 0 \leq \alpha \leq \frac{1}{2}; \\ 0, & \text{otherwise,} \end{cases}$$

where $B_4 = f(\frac{\sqrt{5}+1}{2}, 4\alpha(1-\alpha))$.

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Proof of main result: Theorem 11

- Upper bound ($0 \leq \hat{\nu} \leq \frac{\nu}{1+\nu} < \nu \leq \infty$)

Difficulty: The fractal sets $\mathcal{U}(y, \hat{\nu})$, $E(\hat{\nu})$ and $E(\hat{\nu}, \nu)$ are **not the so-called limsup sets**.

Lemma 1

Let $x \in E(\hat{\nu}, \nu)$, where $\nu > 0$. If the continued fraction expansion of x is not periodic, there exist two ascending sequences $\{n_k\}_{k=1}^{\infty}$ and $\{m_k\}_{k=1}^{\infty}$ depending on x such that:

- (1) $n_k < m_k < n_{k+1} < m_{k+1}$ for $k \geq 1$;
- (2) $a_{n_k+1}(x) = \dots = a_{m_k}(x) = i$ for $k \geq 1$;
- (3) $\liminf_{k \rightarrow \infty} \frac{m_k - n_k}{n_{k+1}} = \hat{\nu}$, $\limsup_{k \rightarrow \infty} \frac{m_k - n_k}{n_k} = \nu$.

Proof of Theorem 11: Upper bound

- We collect all points x which satisfy (1), (2) and (3) as in Lemma 1 to form a covering G .
- For $\varepsilon > 0$, we have $\mathcal{H}^{s_{B_1} + \varepsilon}(G) < +\infty$.

$\Rightarrow \dim_H E(\hat{\nu}, \nu) \leq s_{B_1}$ by the arbitrariness of ε .

Proof of main result: Theorem 11

Lower bound

- Construct a Cantor subset $E(B)$ of $E(\hat{\nu}, \nu)$.
- Estimate the dimension of $E(B)$ by the mass distribution principle.
- Build an injective mapping f from $E(B)$ to $E(\hat{\nu}, \nu)$ and prove that f is dimension-preserving.

Construction of the set $E(B)$

We first define two sequences $\{n_k\}, \{m_k\}$ as follows:

- (1) $n_k < m_k < n_{k+1}$ and $(m_k - n_k) \leq (m_{k+1} - n_{k+1})$ for $k \geq 1$;
- (2) $\lim_{k \rightarrow \infty} \frac{m_k - n_k}{n_{k+1}} = \hat{\nu}$;
- (3) $\lim_{k \rightarrow \infty} \frac{m_k - n_k}{n_k} = \nu$.

Then, we define

$$E(B) = \left\{ x \in [0, 1) : a_n(x) \begin{cases} \in [1, B], & n_k < n \leq m_k; \\ = i, & \text{otherwise.} \end{cases} \right\}.$$

Thanks

Thanks for your attention!